# Lecture 1

## Heaviside / Maxwell

$$\nabla \cdot \vec{E} = \frac{\rho_v}{\varepsilon_0}$$ ← volume charge density → Gauss Law

$\varepsilon_0 = 8.85\ pF/m$ → capacitance per meter of the "universe".

$$\nabla \cdot \vec{H} = 0$$ → Gauss's law ⇒ $\nabla \cdot \vec{B} = 0$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$ → Faraday's law

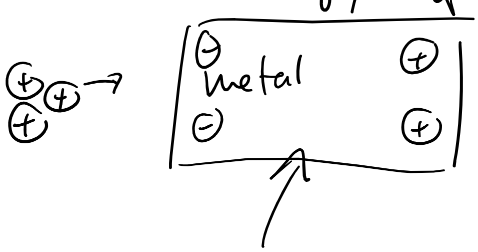
* $$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}$$ → Ampere's Law ; $\mu_0 = 4\pi \times 10^{-7}\ \frac{H}{m}$

$\mu_0$: inductance per distance of the universe

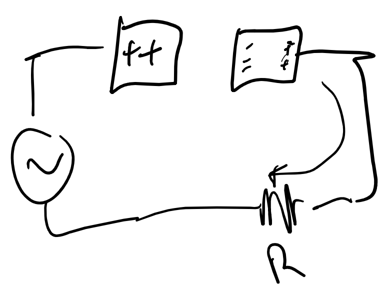
## Magnetic fields

$\vec{B}$ (magnetic flux density [Tesla's]) $= \mu_0 \vec{H}$ (magnetic flux intensity)

$\vec{D}$ (electric flux density / Displacement) $= \varepsilon_0 \vec{E}$ (electric flux density)




Act of Rearranging the charge in a metal is called displacement.



## Wave Eq.

Math: General Diff. Eq.

No sources

$$\frac{\partial^2 \vec{X}^2}{\partial t^2} - |\vec{v}|^2 \vec{X}^2 = 0$$

velocity

$\vec{X} \rightarrow$ is some vector field

$v$ = velocity

$V$ = volume

$V$ = Voltage

## Derive Electric wave (free space) → vacuum

cross sectional area ($S$)

$$\nabla \times (\nabla \times \vec{E}) = \cancel{\nabla}\left(\frac{\partial \vec{B}}{\partial t}\right) \rightarrow V = -N\frac{d\Psi}{dt} \rightarrow V = -N\frac{d(B,S)}{dt}$$

$$\nabla(\cancel{\nabla \cdot \vec{E}}^{\,0}) - \nabla^2 \vec{E} = -\frac{\partial}{\partial t}(\nabla \times \vec{B})$$

$$-\nabla^2 \vec{E} = -\frac{\partial}{\partial t}\left(\cancel{\mu_0 \vec{J}}^{\,c} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}\right)$$

$$\cancel{\vec{J}_c}^{\,c} + \vec{J}_D$$

$$\nabla^2 \vec{E} = \frac{\partial}{\partial t}\left(\mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t}\right) \rightarrow \mu_0 \varepsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\frac{\partial^2 \vec{E}}{\partial t^2} - \left(\frac{1}{\mu_0 \varepsilon_0}\right) \nabla^2 \vec{E} = 0$$

$$\frac{1}{\mu_0 \varepsilon_0} = v^2$$

$$v = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}$$

$$\boxed{v = c = \text{speed of light } 3 \times 10^8 \text{ m/s}}$$

Magnetic wave

$$\frac{\partial^2 \vec{B}}{\partial t^2} - \frac{1}{\mu_0 \varepsilon_0} \nabla^2 \vec{B} = 0$$